

REQUIREMENTS DOCUMENT

COE 454 – Software Engineering II

Client Project Requirements Document

Project Information

Team Name	Team Xenon
Project Codename	Quick Start Pro
Client	Quick Start Driving School
Sector	Education (Driver Training)
Location	Ayeduase Gate, Kumasi
Prepared By	Team Xenon
Date	11/07/2026

Table of Contents

- 1. Business Context
- 2. Problem Statement
- 3. User Personas
- 4. Functional Requirements
- 5. Non-Functional Requirements
- 6. Out of Scope
- 7. Open Questions
- 8. Appendix

1. Business Context

Quick Start Driving School is a privately operated driving school located at Ayeduase Gate, Kumasi. The school provides theoretical and practical driving lessons to individuals seeking driver's license certification. Its services include student registration, scheduling of lessons, payment collection, expenditure tracking, and record management.

The school currently serves approximately 15-20 students each month and is run by a Manager, a Secretary, and a team of Driving Instructors.

Most administrative operations — including cash-flow registration of income and expenditure, student registration, and scheduling — are still handled manually, which makes day-to-day work tedious and reduces transparency and accountability. The client has indicated a clear need for a software system to bring these processes under proper control.

Notably, the client previously attempted to adopt a software system to digitise parts of its operations. That system was ultimately abandoned due to Identity and Access Management (IAM) shortcomings — users were able to access and edit information beyond their role, including vital records, with no reliable way to track who made a change. This history directly informs the access-control and audit-logging requirements captured in this document, and the team will take care not to repeat the same shortcomings.

In a follow-up meeting with the owner (July 2026), the team heard the failure recounted first-hand: a previous secretary initially edited saved records in order to reprint receipts, and when editing was later restricted, deleted entire student profiles and recreated them so that fresh receipts could be issued. This is why payment records in the new system must lock once a receipt has been issued, and why deletions must be captured in a tamper-proof audit log.

The same meeting clarified two structural points. First, every student enrolls for Driving only, Licence only, or Driving + Licence, and licence processing applies only to licence-inclusive enrolments. Second, course duration is inherently variable because attendance frequency differs per student, so packages are defined by lesson count — typically 15 practical lessons — rather than by fixed duration.

The primary stakeholders include:

- Business Owner / Manager
- Secretary (Administrative Staff)
- Driving Instructors
- Students

2. Problem Statement

During our site visits, we observed that Quick Start Driving School relies heavily on manual processes for its day-to-day operations.

Cash-flow registration — recording budgets, income, and expenditures — is performed manually, which makes the work tedious and significantly reduces transparency and accountability. Student registration and scheduling are also handled on paper, and payment balances must be recalculated by hand whenever a new payment is made, which is slow and error-prone.

In addition, the client's previous experience with a digital system revealed a further problem: without proper Identity and Access Management, staff were able to view and edit information outside their role, including vital business and student records, and there was no reliable log of who changed what. This eroded trust in that system and is a key reason it was discontinued.

As student enrollment increases, retrieving information becomes slower, duplicate entries occur, and important records may be misplaced or altered without oversight.

These challenges lead to:

- Tedious, error-prone cash-flow tracking
- Poor transparency and accountability over finances
- Slow student registration and scheduling
- Unauthorized or untracked edits to vital records
- Manual recalculation errors in payments and balances
- Limited visibility into schedules for instructors and staff

The business requires a secure, role-based software system that streamlines these operations while giving the Manager confidence that vital data is protected and every change is accountable.

3. User Personas

Persona 1 – Manager / Business Owner

Name	Reagan Masud
Role	Manager / Business Owner
Responsibilities	• Supervise day-to-day operations • Oversee Secretary and Driving Instructors • Monitor financial records and cash flow • Approve expenditures • Manage staff access and permissions
Goals	• Reliable, role-based access control • Full visibility into schedules and finances • Accountability for every change made in the system • Reduce manual, paper-based work
Frustrations	• Identity Access Management (IAM) issues experienced with a previous software system • Inability to tell who edited a given record • Difficulty monitoring cash flow manually

Persona 2 – Secretary

Name	Mercy Otoo
Role	Secretary / Administrative Staff
Responsibilities	• Keep student and business records • Register new students • Manage student scheduling • Record and track payments
Goals	• Register students quickly without paper forms • Avoid manually recalculating payments and balances • Keep an accurate, searchable schedule • Reduce duplicate or lost records
Frustrations	• Manual, paper-based registration • Recalculating payments and balances by hand

	• Time lost searching for records
--	---

Persona 3 – Driving Instructor

Name	Solomon Adomako and Obed Owusu Opoku
Role	Driving Instructor
Responsibilities	• Deliver theory lessons • Conduct practical driving sessions • Track student attendance and progress
Goals	• Clear, up-to-date view of assigned schedule • Easy way to record attendance and progress
Frustrations	• None reported at this time

Note (July 2026): Driving Instructors will not log into the system in Phase 1. Attendance and lesson records are entered by the Secretary, who records which instructor took each lesson. Instructor-facing stories (e.g., User Story 6) are deferred to Phase 2.

4. Functional Requirements

The following user stories capture the core system features requested by the client: Identity and Access Management, data protection and integrity, financial and admin record-keeping, scheduling, attendance, and payment tracking.

User Stories 13–16, the Client Priorities subsection, and the amendments to Stories 4–12 below were added following the requirements meeting held with the business owner in July 2026.

User Story 1

As a Manager

I want to log into the system using secure, role-based credentials

So that only authorized personnel can access sensitive business data.

Acceptance Criteria:

- Login requires a unique username and password.
- Each account is assigned exactly one role: Manager and Secretary.
- Failed login attempts are logged.

User Story 2

As a Manager

I want to restrict which fields and records each role is allowed to edit

So that vital business and student information cannot be altered by unauthorized users, unlike the team's experience with a previous system.

Acceptance Criteria:

- Role-based permissions govern edit access to every record type.
- Attempts to edit a restricted field are blocked and logged.
- Only the Manager can grant or revoke edit permissions.

User Story 3

As a Secretary

I want to register new students directly in the system

So that I no longer need to fill out and file paper registration forms.

Acceptance Criteria:

- Student name and phone number are required fields.
- Registration date is recorded automatically.
- Duplicate student entries are flagged before saving.

User Story 4

As a Secretary

I want to have the system automatically calculate payments and outstanding balances

So that I do not have to manually recalculate figures for every student.

Acceptance Criteria:

- Outstanding balance updates automatically after each payment.
- A receipt number is generated for every payment.
- Payment history is retained per student.
- Once a payment is saved and its receipt issued, the record is locked: the Secretary cannot edit or delete it, and corrections require the Manager.

User Story 5

As a Secretary

I want to create and manage lesson schedules for students and instructors

So that lesson times are tracked accurately and conflicts are avoided.

Acceptance Criteria:

- A schedule entry can be created per student and per instructor.
- Overlapping schedule entries are flagged.
- Schedule changes are saved immediately and logged.
- Each schedule slot displays the number of participants booked into it.
- Theory and practical sessions are clearly distinguished.

User Story 6

As a secretary

I want to record student attendance when each student signs in for a lesson

So that participation can be tracked accurately over time.

Acceptance Criteria:

- Attendance is recorded by sign-in only; no sign-out is required, per the client's current workflow.
- Attendance history is viewable per student.
- Repeated absences are flagged for follow-up.

User Story 7**As a Secretary**

I want to track a student's course progress

So that the Manager can see lessons taken and lessons remaining at a glance.

Acceptance Criteria:

- Progress is measured as lessons attended versus lessons included in the student's package.
- Lessons taken and lessons remaining are visible on the student profile.
- Optional remarks can be added when a lesson is marked done.

User Story 8**As a Manager**

I want to view financial records covering income and expenditure

So that I can monitor cash flow and business performance.

Acceptance Criteria:

- Income and expenditure are summarized by period.
- Records can be filtered by date range.
- Totals are calculated automatically.
- Expenditure categories are predefined regular items (e.g., fuel, learner licence fees, vehicle maintenance), with the full list finalized in consultation with the Secretary.
- Fuel and maintenance expenses can be tagged to a specific vehicle across the school's three cars.

User Story 9**As a Manager**

I want to view an admin log of changes made in the system

So that I can hold staff accountable for the actions they take.

Acceptance Criteria:

- Every change is logged with user, action, and timestamp.
- Logs are read-only and cannot be edited or deleted by regular users.
- Logs can be filtered by user, role, or date.

User Story 10**As a Secretary**

I want to maintain a centralized record for every student

So that personal details, payments, schedule, and progress are all in one place.

Acceptance Criteria:

- A student profile stores contact info, payment history, schedule, and progress.
- Profiles are searchable by name or phone number.
- No duplicate student profiles are permitted.
- The profile opens with an at-a-glance summary: registration date, enrolment type, total paid and outstanding balance, lessons taken and remaining, and licence status.
- All profile fields remain editable by the Secretary except locked payment records.

User Story 11

As a Manager

I want to have the system generate reminders for upcoming lessons and outstanding payments
So that staff and students stay informed without manual follow-up.

Acceptance Criteria:

- Reminders are generated for upcoming scheduled sessions.
- Reminders are generated for outstanding balances.
- Reminders are generated for upcoming scheduled licence examination dates.
- Delivery channel for reminders is configurable (see Open Questions).

User Story 12

As a Secretary

I want to track each licence-inclusive student's licensing process through its defined stages
So that the school always knows exactly where every student stands in the licence pipeline.

Acceptance Criteria:

- The tracked stages are: eye test done; learner licence issued / not issued; scheduled examination date; examination result (passed / failed); licence issued.
- A future examination date can be entered in advance so that it is not forgotten.
- The student's current licence status is visible on the profile at a glance.
- Licence tracking applies only to students whose enrolment includes licensing.

User Story 13

As a Manager

I want to generate a list of students whose learner licences have or have not been issued
So that I can compile DVLA batches and follow up without requesting a manual report.

Acceptance Criteria:

- The list can be filtered by licence status (issued / not issued).
- Summary counts are displayed (e.g., 100 registered, 20 learner licences pending).
- When a learner licence is marked issued, the student immediately moves to the issued list.

User Story 14

As a Secretary

I want to record which driving instructor took each completed lesson
So that the Manager can see how many lessons each instructor has delivered.

Acceptance Criteria:

- The instructor is selected with a single tick when a lesson is marked done.
- Totals per instructor are available (e.g., total 100 lessons — Instructor A 40, Instructor B 20).
- Totals can be filtered by week or by any date range for comparison across periods.

User Story 15

As a Secretary

I want to classify each student at registration as Driving only, Licence only, or Driving + Licence
So that the system shows each student only the sections relevant to their enrolment.

Acceptance Criteria:

- Enrolment type is a required field at registration.
- Licence tracking appears only for licence-inclusive enrolments.

- The enrolment type is displayed in the profile's at-a-glance summary.

Client Priorities (July 2026 meeting)

When asked to rank the requested features against the limited project timeline, the owner was explicit about what matters most:

- Top priority: student registration, payment tracking with automatic balances, and an at-a-glance student profile showing payments, lessons taken, and licence status.
- Scheduling should remain lean: participant counts per slot and which instructor took each lesson matter more than precise timing.
- Reporting should not be over-engineered; the reports named under Open Questions (Financial Aspects) are sufficient.
- Functionality over aesthetics: the system will be judged on the problems it solves, not on visual polish.

5. Non-Functional Requirements

Security

Given the client's past issues with Identity and Access Management, security is a priority requirement, not an afterthought.

- Only authenticated users may access the system.
- Access is role-based: Manager, Secretary, and Instructor each see only what their role permits.
- Passwords are securely encrypted (never stored in plain text).
- All edits to vital records are logged with user, timestamp, and prior value.

Data Integrity

- No duplicate student records may exist.
- Vital fields (e.g., payment amounts, balances) cannot be edited by roles without explicit permission.
- Input validation prevents malformed or incomplete records from being saved.

Performance

- Pages should load within 3 seconds under normal internet conditions.

Availability

- The system should be available at least 99% of the time during business hours.

Usability

- Staff with basic computer literacy should be able to operate the system with minimal training.

Responsiveness

- The application should work properly on desktops, tablets, and smartphones.

Backup

- Business, financial, and student data should be backed up on a regular schedule.

6. Out of Scope

The following are proposed exclusions from Phase 1 (MVP). Items marked (to confirm) should be verified with the client before the requirements are finalized.

- Printed/physical receipts system — record-keeping will be digital-first; the client currently uses physical receipt books, and this workflow is being replaced rather than replicated.
- Online payment integration (e.g., MTN MoMo/Paystack) — confirmed as Phase 2 (July 2026): the client is open to it and already operates a payment platform that routes funds directly to the business account.
- Student-facing mobile application or self-service portal (to confirm)
- Automated SMS notifications — reminders will initially be in-system only (to confirm)
- Government/DVLA examination integration (to confirm)

7. Open Questions

Questions resolved during the July 2026 meeting with the owner are marked “Answered” below; the remainder will be clarified during subsequent meetings.

Financial Aspects

- What specific financial reports does the Manager require (daily, weekly, monthly)? — **Answered (Jul 2026): expenditure reports, lessons-per-instructor totals, and licence status lists; reporting beyond these should stay minimal.**
- Should expenditure categories be predefined by the team or customizable by the client? — **Answered (Jul 2026): predefined regular categories (fuel, learner licence fees, vehicle maintenance, among others), finalized with the Secretary; fuel is tracked per vehicle.**
- Should receipts be generated in-system even though physical printing is out of scope? — **Answered (Jul 2026): yes — every payment generates an in-system receipt with a unique number.**

Access Control

- What specific transactions or fields should be restricted per role (Manager, Secretary, Instructor)? — **Answered in part (Jul 2026): payment records lock once a receipt is issued; all other profile fields remain editable by the Secretary.**
- Should the Secretary be able to edit a payment record after it has been saved, or only the Manager? — **Answered (Jul 2026): no — once saved and receipted, only the Manager can correct a payment.**
- What access level, if any, should Driving Instructors have to financial data? — **Answered (Jul 2026): none; instructors do not access the system in Phase 1.**
- What specifically went wrong with the previous system's IAM setup, so the team can be sure not to repeat it? — **Answered (Jul 2026): see Business Context — receipts were edited, and profiles were deleted and recreated to reissue them.**
- Should edits to vital records require a Manager approval step, and should edit history be recoverable? — **Answered (Jul 2026): yes — corrections to locked records go through the Manager, and the audit log retains prior values.**

Schedule Monitoring

- Should schedule conflicts be automatically prevented, or simply flagged for manual resolution? — **Answered (Jul 2026): flagging is sufficient; the priority is showing participant counts per slot.**
- Should students be able to view (or request) their own schedule, or is this admin-only?

General

- How many staff members (Secretaries/Instructors) will use the system?
- Will the system be hosted online or locally, and what is the estimated hosting budget?
- How long should student records be retained after course completion?

8. Appendix

Assumptions

- The business has stable internet connectivity.
- Staff possess basic smartphone/computer skills.
- The school is willing to fully transition from paper records to a digital system.
- The system will initially be used by one Manager, one Secretary, and multiple Instructors.

Constraints

- Limited project duration
- Budget constraints for hosting and maintenance.
- Initial deployment targets core business processes only.
- The system must avoid repeating the Identity Access Management shortcomings of the client's previous software.